Story: the PCLAI widget

> *"Where, in PCA-derived ancestry space, does each haplotype's point cloud spread — and where are the breakpoints between its ancestral segments?"*

The PCLAI widget is the most ambitious of the three. It is the only one that drives **two coupled visualizations at once**: it emphasizes nodes in the 3D graph *and* steers a separate 2D chart. The widget's list is the gesture surface; the graph and the chart are two synchronized views of what that gesture means.

PCLAI ("Point Cloud Local Ancestry Inference"; Geleta et al., *Nature Genetics* 2026) is a continuous alternative to discrete local ancestry inference. Instead of labeling each genomic window with one of a few ancestry categories, PCLAI assigns each window a **coordinate** in a genetic embedding space — by default the **PCA space** of a labeled reference panel, where Euclidean distance approximates the F_2 genetic distance. An individual haplotype is therefore not a string of labels but a **point cloud**: one point per genomic window, scattered through PCA space according to local ancestry.

In PGB, each graph node represents one such genomic-window-aligned haplotypic segment. So every node carries a PCA coordinate per assembly-haplotype that walks through it — exactly the per-window coordinates the PCLAI model produces. The boundaries between graph nodes are, structurally, the recombination **breakpoints** the model regresses between.

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1. The widget

Open the PCLAI card and you see:

- A **search field**.
- A **scrollable list of coordinate keys** — strings of the form `ASM#HAP` (e.g. `HG00408#1`). Visually it looks identical to the Assembly list, but each row here is a *placement-bearing identity*, not just an assembly identity.

The gesture is binary again: zero or one coordinate key is selected.

But the widget has a second behavior the other two do not: just by being **open**, it activates an **absence** mode. Every node that lacks PCLAI placement data is marked with a distinct neutral color from the moment the card appears — before the user has selected anything. The mere act of opening the widget reveals where the placement model could not speak. (This is bookkeeping that survives card close/reopen via `pclaiAbsenceCoordinator`.)

`src/widgets/pclaiWidget.ts`

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2. What the widget is reading

The list is populated from `pclaiCoordinateService.getAllCoordinateKeys()` — the union of every coordinate key that appears anywhere in any node's `pclai_hg38` or `pclai_asm` block. The service builds, at load time, a dense set of indexes from the dataset's per-node placement data:

Index	Shape	Used by
`coordinatesForNode`	`nodeId → Map<coordKey, {coords, RGB}>`	chart, on hover
`coordinatesForCoordinateKey`	`coordKey → Map<nodeId, {coords, RGB}>`	chart, on selection
`nodeIdsWithCoordinateKey`	`coordKey → Set<nodeId>`	3D graph emphasis
`absentNodeSet`	`Set<nodeId>` (nodes with `pclai_hg38 == {}`)	absence coloring
per-key color maps	`coordKey → (nodeId → RGB)`	3D graph color

Like the assembly story, the *data-as-stored* shape (per-node lists of placements) is inverted at ingest into a *data-as-used* shape (per-key lookups). Without these indexes, every gesture would scan the entire dataset.

The colors in PCLAI are not chosen — **they ship with the data**, in the `RGB` field next to each coordinate. They are the model's own visual encoding of each point's PCA location (so that two points close in PCA space are also close in color); the visualization reads them, it does not interpret.

Each node carries two PCLAI blocks side by side — `pclai_hg38` and `pclai_asm` — distinguished by their `pclai_coord_system` field (`"GRCh38"` vs `"assembly"`). The two indicate which coordinate system the window was anchored in; both produce a `[x, y]` PCA placement, an `RGB`, and a `confidence_score`. Either may be empty (`{}`) for a given node-assembly pair when the model has no placement to report.

`src/widgets/pclaiCoordinateService.js`

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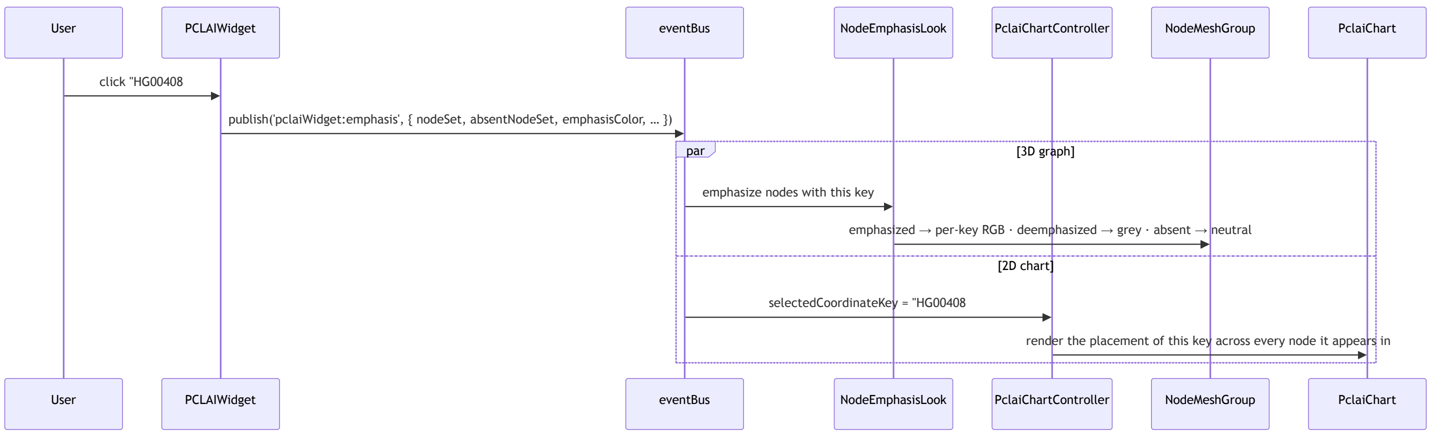
3. What the scene does in response

A single selection drives two listeners in lockstep — the 3D graph and the 2D chart.



The chart's state machine

The PCLAI chart is a 2D scatter of the PCA space. Its constant backdrop is the **reference panel** — the labeled populations PCLAI was trained against (loaded from `hprc-reference-pca.tsv`), drawn as colored dots that visually divide the space into AFR / EUR / EAS / AMR / WAS regions. Every other dot the chart draws is read against this backdrop.

The controller renders the chart as a **pure function** of two pieces of state:

`hovered` (a node)	`selected` (a coord key)	Chart shows
∅	∅	**Idle** — reference panel only, full color
set	∅	One genomic window, **every haplotype's placement** at that window
∅	set	One haplotype, **its full point cloud** across every node it walks
set	set	The **single** point: this haplotype at this node

The hovered node comes from a graph-side raycaster firing `lineIntersection`; the selected coord key comes from a widget click. Crossing the two is what makes the chart feel coupled to the graph.

`src/widgets/pclaiChartController.js` · `src/widgets/pclaiChart.js`

The three node colorings, simultaneously

In emphasis mode the graph's nodes are painted in three categories at once:

- **Emphasized** — nodes that carry the selected coord key, painted in the per-node RGB the data ships with.
- **Deemphasized** — nodes that have PCLAI placements but not for this key, painted muted grey.
- **Absent** — nodes that have no PCLAI placement at all, painted in a separate neutral tone.

The absence color is reserved — it stays even with no selection (whenever the widget is open), so the user always knows which nodes the model is *silent* about. This is what makes PCLAI's third category necessary; neither Assembly nor Population needs to express "no data."

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4. What the scientist is doing

PCLAI supports the **placement gesture**: rather than asking *which individuals* (Assembly) or *which populations* (Population), the scientist asks *where in PCA space* each haplotype's genomic segments sit, and reads admixture and recombination structure directly off the cloud's geometry. The paper's central observation is that the **shape and spread of a haplotype's point cloud (Tr(Σ), the trace of its covariance) is itself the admixture signal** — tight clusters mean single-ancestry haplotypes; broad, multi-modal clouds mean admixed ones.

Three readings the gesture supports:

- **Hover a node, see a cross-section.** With no selection, hovering a node populates the chart with every haplotype's placement at that single genomic window. A tight cluster within one reference region means most individuals trace this segment to the same ancestry; a spread-out cloud means the locus is itself a site of high inter-individual variation.
- **Select a haplotype, see its point cloud.** With one coord key selected, the chart shows the haplotype's *entire* point cloud — one dot per node it walks. A tight cluster inside one reference region (e.g. EUR) means this individual's ancestry is largely from there. Multiple separated clusters (EUR + AMR, say) reveal admixture, and the visible cluster sizes are proportional to how much of the genome traces to each. An elongated single axis suggests admixture along one drift direction.
- **Hover and select together, see one point.** The intersection: where does *this specific haplotype* place at *this specific genomic window*? A single dot, or nothing if the model has no opinion.

The simultaneous 3D-graph emphasis grounds every chart reading spatially: the chart says *where in PCA space*, the graph says *where on the genome*. When the same individual's PCA dots hop between reference regions as you move along the graph, the node boundaries where the hop happens are the candidate **recombination breakpoints**.

where the hop happens are the candidate **recombination breakpoints** PCLAI is built to detect. Neither view alone tells that story.

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5. What this story doesn't tell you

- **Which individuals merely *visit* a node** — the Assembly widget covers presence-on-graph; PCLAI covers *placement-in-learned-space*, which is a much stronger claim.
- **Aggregate distribution by ancestry group** — the Population widget gives the demographic landscape view, without per-individual coordinates.
- **Why the model is silent on some nodes** (the absent set) — that is the heart of Issue #77, still open.

The PCLAI widget answers two questions at once — *"where in PCA space?"* and *"with which recombination structure?"* — by binding a list selection to a graph emphasis and a chart render in the same gesture.

Pointers

- Widget: `src/widgets/pclaiWidget.ts`
- Service: `src/widgets/pclaiCoordinateService.js`
- 3D look: `src/looks/nodeEmphasisLook.ts` (shared with Assembly)
- Chart: `src/widgets/pclaiChart.js` · `src/widgets/pclaiCoordinateSpace.js`
- Chart controller (state machine): `src/widgets/pclaiChartController.js`
- Absence coordinator: `src/widgets/pclaiAbsenceCoordinator.js`
- Data shape: `pclai_hg38` and `pclai_asm` blocks per node — see `dataset-skeleton.html`.
- Reference paper: Geleta, Bu, Turner, et al. *Point cloud local ancestry inference (PCLAI): continuous coordinate-based ancestry along the genome.* Nature Genetics, 2026. (`~/PanGenomeProject/pclai-nature-paper-2026/`)
- Reference panel: `hprc-reference-pca.tsv` — the labeled PCA backdrop for the chart.